

Auteur: Stan Van Campenhout
Studierichting: Informatica
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$$\begin{aligned} \int_0^{\frac{\pi}{2}} \frac{1}{2 + \sin x} dx &\stackrel{t=\tan(\frac{x}{2})}{=} \int_0^1 \frac{1}{1 + t^2 + t} dt = \int_0^1 \frac{1}{\left(t + \frac{1}{2}\right)^2 + \frac{3}{4}} dt \\ &= \left[\frac{2}{\sqrt{3}} \operatorname{Bgtan} \left(\frac{2t + 1}{\sqrt{3}} \right) \right]_0^1 = \frac{2\pi}{3\sqrt{3}} - \frac{\pi}{3\sqrt{3}} = \frac{\pi}{3\sqrt{3}} \end{aligned}$$

References